

A Timestamp Audit of Filing-Tone Signals

A known-truth worked final-project example

FIN 7057

2026-09-27

The research question

Company filings carry **tone**. The hopeful pitch: a contextual reader extracts more useful information than a word list.

The disciplined question is narrower:

*Does a richer filing-tone reader predict the drift **after the filing is public**, better than a simple lexicon, **once we respect the timestamp and pay costs**?*

Everything hinges on those last two conditions.

Prior evidence narrows the question

- ▶ Finance-specific dictionaries improve on generic word lists (Loughran and McDonald 2011).
- ▶ Finance-adapted contextual models improve several sentiment-classification tasks (Araci 2019; Huang, Wang, and Yang 2023).
- ▶ Filing-language changes have been associated with later firm outcomes and returns (Cohen, Malloy, and Nguyen 2020).

Unresolved here: does a richer text score improve a feasible downstream return-ranking decision relative to the simple finance baseline?

The data (and the planted truth)

Reproducible **synthetic filings** (a stand-in for SEC EDGAR — runs offline, fixed seed). Each filing has a latent tone, and two returns follow it:

- ▶ an **announcement** jump at the filing date — strongly tone-driven, and *untradeable* before the filing is public;
- ▶ a faint **post-filing drift** over the next 20 days — weakly tone-driven, and the only genuinely tradeable target.

The filing is dated **45 days after** the period it describes. That gap is the central information-clock risk.

Two readers of the text

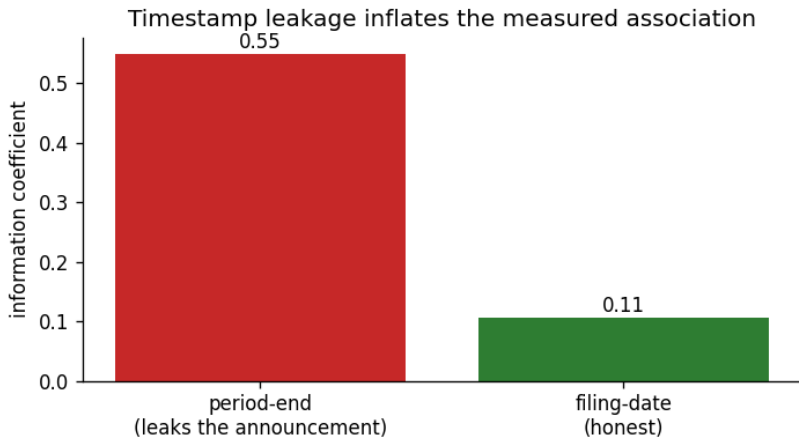
- ▶ **Baseline** — a shallow Loughran–McDonald-style lexicon: net (positive – negative) hits. Cheap, transparent, hard to beat.
- ▶ **Candidate** — a richer contextual-reader proxy. The pipeline can call **FinBERT**, but the committed offline output uses a deterministic stand-in.

The candidate must improve on the baseline out of sample before its additional complexity counts for anything.

The leakage trap

Align tone to the **period-end** date and the return window swallows the announcement pop — a move you could never have traded.

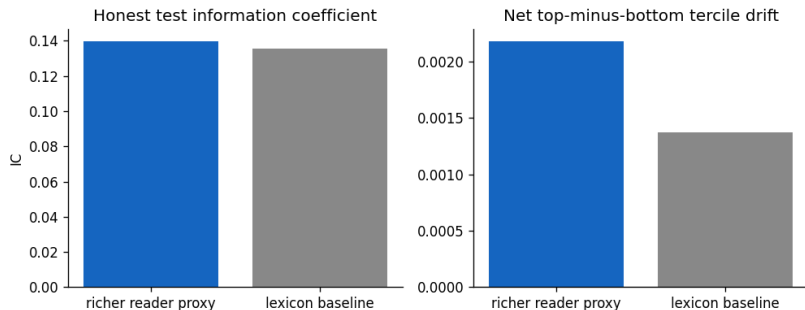
Align to the **filing date** and you keep only the honest, post-public drift.



The same score and text produces roughly **five times the**

Honest, time-aware evaluation

Evaluate out of sample (never shuffle time). Compare the **information coefficient** and a top-minus-bottom **tercile spread**, net of a 10 bp cost.



The result

- ▶ Removing the timestamp leak cuts the apparent signal by $\sim 5\times$.
- ▶ On the honest target, the richer reader's edge is **small and does not clearly beat the lexicon**.
- ▶ The experiment does not provide enough uncertainty evidence to treat that small difference as stable.

This is a good negative result — the disciplined finding is that the easy “NLP alpha” was mostly leakage.

Limitations

- ▶ **Synthetic data** — the experiment does not estimate predictability in real SEC filings.
- ▶ **Offline proxy** — the committed run does not test FinBERT itself.
- ▶ **One universe, one horizon** — no sector/size controls, a single tercile cut.
- ▶ **Stylized 10 bp cost** — a filing-clustered strategy would likely pay more impact.
- ▶ **No multiple-testing accounting** — a real model/lexicon/horizon search needs a deflated Sharpe and a trial count.

Takeaway

The question was never *“does the model sound fluent?”*

It is *“does its tone predict a return you could actually have traded, after costs?”*

Here — **barely**. And that answer, honestly arrived at, is the assignment.

References

- Araci, Doğu. 2019. "FinBERT: Financial Sentiment Analysis with Pre-Trained Language Models." *arXiv Preprint arXiv:1908.10063*. <https://doi.org/10.48550/arXiv.1908.10063>.
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